

TestAmerica

THE LEADER IN ENVIRONMENTAL TESTING

ANALYTICAL REPORT

TestAmerica Laboratories, Inc.

TestAmerica Michigan

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TestAmerica Job ID: 190-17646-1

Client Project/Site: City of Cadillac

For:

City of Cadillac Utilities

1121 Plett Road

Cadillac, Michigan 49601

Attn: Cindy Tomaszewski

Sue Schafer

Authorized for release by:

10/29/2018 7:51:04 AM

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Results relate only to the items tested and the sample(s) as received by the laboratory.



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Sample Summary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac

TestAmerica Job ID: 190-17646-1

Lab Sample ID	Client Sample ID	Matrix	Collected	Received
190-17646-1	Leachate	Water	10/03/18 09:30	10/04/18 12:41
190-17646-2	Leachate (duplicate)	Water	10/03/18 09:35	10/04/18 12:41
190-17646-3	Equipment Blank	Water	10/03/18 09:25	10/04/18 12:41
190-17646-4	Field Blank	Water	10/03/18 00:00	10/04/18 12:41

Case Narrative

Client: City of Cadillac Utilities
Project/Site: City of Cadillac

TestAmerica Job ID: 190-17646-1

Job ID: 190-17646-1

Laboratory: TestAmerica Michigan

Narrative

Job Narrative 190-17646-1

Comments

No additional comments.

Receipt

The samples were received on 10/4/2018 12:41 PM; the samples arrived in good condition, properly preserved and, where required, on ice. The temperature of the cooler at receipt was 0.0° C.

LCMS

Method(s) 537 (modified): Results for samples Leachate (190-17646-1) and Leachate (duplicate) (190-17646-2) were reported from the analysis of a diluted extract due to matrix interferences in the analysis of the undiluted extract. The dilution factor was applied to the labeled internal standard area counts and these area counts were within acceptance limits

Method(s) 537 (modified): Results for samples Leachate (190-17646-1) were reported from the analysis of a diluted extract due to matrix interferences in the analysis of the undiluted extract. The dilution factor was applied to the labeled internal standard area counts and these area counts were within acceptance limits.

Method(s) 537 (modified): The following samples were diluted due to the nature of the sample matrix: Leachate (190-17646-1) and Leachate (duplicate) (190-17646-2). Elevated reporting limits (RLs) are provided.

Method(s) 537 (modified): The Isotope Dilution Analyte (IDA) recovery for Perfluorotetradecanoic acid (PFTeA) associated with the following samples is below the method recommended limit: Leachate (190-17646-1) and Leachate (duplicate) (190-17646-2). This sample was re-analyzed at greater dilution with concurring results. Generally, data quality is not considered affected if the IDA signal-to-noise ratio is greater than 10:1, which is achieved for all IDA in the samples.

Method(s) 537 (modified): Results for samples Leachate (duplicate) (190-17646-2) were reported from the analysis of a diluted extract due to matrix interferences in the analysis of the undiluted extract. The dilution factor was applied to the labeled internal standard area counts and these area counts were outside acceptance limits. The internal standard is not used to quantitate target analytes; therefore, there is no impact to the data.

No additional analytical or quality issues were noted, other than those described above or in the Definitions/Glossary page.

Organic Prep

Method(s) 3535: The following samples are a brown color prior to extraction:
Leachate (190-17646-1) and Leachate (duplicate) (190-17646-2).

Method Code: 3535 PFC
preparation batch 320-252552

Method(s) 3535: The following samples are a yellow color after extraction:
Leachate (190-17646-1) and Leachate (duplicate) (190-17646-2).

Method Code: 3535 PFC
preparation batch 320-252552

No additional analytical or quality issues were noted, other than those described above or in the Definitions/Glossary page.

Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac

TestAmerica Job ID: 190-17646-1

Client Sample ID: Leachate

Date Collected: 10/03/18 09:30

Date Received: 10/04/18 12:41

Lab Sample ID: 190-17646-1

Matrix: Water

Method: 537 (modified) - Fluorinated Alkyl Substances

Analyte	Result	Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
Perfluorobutanoic acid (PFBA)	950		21	ng/L		10/16/18 08:36	10/25/18 04:26	10
Perfluoropentanoic acid (PFPeA)	610		21	ng/L		10/16/18 08:36	10/25/18 04:26	10
Perfluorohexanoic acid (PFHxA)	2100		21	ng/L		10/16/18 08:36	10/25/18 04:26	10
Perfluoroheptanoic acid (PFHpA)	240		21	ng/L		10/16/18 08:36	10/25/18 04:26	10
Perfluorooctanoic acid (PFOA)	590		21	ng/L		10/16/18 08:36	10/25/18 04:26	10
Perfluorononanoic acid (PFNA)	<21		21	ng/L		10/16/18 08:36	10/25/18 04:26	10
Perfluorodecanoic acid (PFDA)	<21		21	ng/L		10/16/18 08:36	10/25/18 04:26	10
Perfluoroundecanoic acid (PFUnA)	<21		21	ng/L		10/16/18 08:36	10/25/18 04:26	10
Perfluorododecanoic acid (PFDoA)	<21		21	ng/L		10/16/18 08:36	10/25/18 04:26	10
Perfluorotridecanoic acid (PFTriA)	<21		21	ng/L		10/16/18 08:36	10/25/18 04:26	10
Perfluorotetradecanoic acid (PFTeA)	<21		21	ng/L		10/16/18 08:36	10/25/18 04:26	10
Perfluorobutanesulfonic acid (PFBS)	950		21	ng/L		10/16/18 08:36	10/25/18 04:26	10
Perfluoropentanesulfonic acid (PFPeS)	160		21	ng/L		10/16/18 08:36	10/25/18 04:26	10
Perfluorohexanesulfonic acid (PFHxS)	610		21	ng/L		10/16/18 08:36	10/25/18 04:26	10
Perfluoroheptanesulfonic Acid (PFHpS)	<21		21	ng/L		10/16/18 08:36	10/25/18 04:26	10
Perfluorooctanesulfonic acid (PFOS)	120		21	ng/L		10/16/18 08:36	10/25/18 04:26	10
Perfluorononanesulfonic acid (PFNS)	<21		21	ng/L		10/16/18 08:36	10/25/18 04:26	10
Perfluorodecanesulfonic acid (PFDS)	<21		21	ng/L		10/16/18 08:36	10/25/18 04:26	10
Perfluorooctanesulfonamide (FOSA)	<21		21	ng/L		10/16/18 08:36	10/25/18 04:26	10
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	<210		210	ng/L		10/16/18 08:36	10/25/18 04:26	10
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	<210		210	ng/L		10/16/18 08:36	10/25/18 04:26	10
4:2 FTS	<210		210	ng/L		10/16/18 08:36	10/25/18 04:26	10
8:2 FTS	<210		210	ng/L		10/16/18 08:36	10/25/18 04:26	10
Isotope Dilution	%Recovery	Qualifier	Limits			Prepared	Analyzed	Dil Fac
13C4 PFBA	27		25 - 150			10/16/18 08:36	10/25/18 04:26	10
13C5 PFPeA	58		25 - 150			10/16/18 08:36	10/25/18 04:26	10
13C2 PFHxA	62		25 - 150			10/16/18 08:36	10/25/18 04:26	10
13C4 PFHpA	81		25 - 150			10/16/18 08:36	10/25/18 04:26	10
13C4 PFOA	92		25 - 150			10/16/18 08:36	10/25/18 04:26	10
13C5 PFNA	91		25 - 150			10/16/18 08:36	10/25/18 04:26	10
13C2 PFDA	92		25 - 150			10/16/18 08:36	10/25/18 04:26	10
13C2 PFUnA	85		25 - 150			10/16/18 08:36	10/25/18 04:26	10
13C2 PFDoA	55		25 - 150			10/16/18 08:36	10/25/18 04:26	10
13C2 PFTeDA	15 *		25 - 150			10/16/18 08:36	10/25/18 04:26	10
13C3 PFBS	108		25 - 150			10/16/18 08:36	10/25/18 04:26	10
18O2 PFHxS	84		25 - 150			10/16/18 08:36	10/25/18 04:26	10
13C4 PFOS	91		25 - 150			10/16/18 08:36	10/25/18 04:26	10
13C8 FOSA	80		25 - 150			10/16/18 08:36	10/25/18 04:26	10
d3-NMeFOSAA	92		25 - 150			10/16/18 08:36	10/25/18 04:26	10
d5-NEtFOSAA	91		25 - 150			10/16/18 08:36	10/25/18 04:26	10
M2-8:2 FTS	132		25 - 150			10/16/18 08:36	10/25/18 04:26	10

TestAmerica Michigan

Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac

TestAmerica Job ID: 190-17646-1

Client Sample ID: Leachate

Date Collected: 10/03/18 09:30

Date Received: 10/04/18 12:41

Lab Sample ID: 190-17646-1

Matrix: Water

Method: 537 (modified) - Fluorinated Alkyl Substances - DL

Analyte	Result	Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
6:2 FTS	<2100		2100	ng/L		10/16/18 08:36	10/25/18 04:19	100
Isotope Dilution	%Recovery	Qualifier	Limits			Prepared	Analyzed	Dil Fac
M2-6:2 FTS	114		25 - 150			10/16/18 08:36	10/25/18 04:19	100

Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac

TestAmerica Job ID: 190-17646-1

Client Sample ID: Leachate (duplicate)

Lab Sample ID: 190-17646-2

Date Collected: 10/03/18 09:35

Matrix: Water

Date Received: 10/04/18 12:41

Method: 537 (modified) - Fluorinated Alkyl Substances

Analyte	Result	Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
Perfluoropentanoic acid (PFPeA)	620		18	ng/L		10/16/18 08:36	10/25/18 04:41	10
Perfluorohexanoic acid (PFHxA)	1900		18	ng/L		10/16/18 08:36	10/25/18 04:41	10
Perfluoroheptanoic acid (PFHpA)	260		18	ng/L		10/16/18 08:36	10/25/18 04:41	10
Perfluorooctanoic acid (PFOA)	700		18	ng/L		10/16/18 08:36	10/25/18 04:41	10
Perfluorononanoic acid (PFNA)	18		18	ng/L		10/16/18 08:36	10/25/18 04:41	10
Perfluorodecanoic acid (PFDA)	<18		18	ng/L		10/16/18 08:36	10/25/18 04:41	10
Perfluoroundecanoic acid (PFUnA)	<18		18	ng/L		10/16/18 08:36	10/25/18 04:41	10
Perfluorododecanoic acid (PFDoA)	<18		18	ng/L		10/16/18 08:36	10/25/18 04:41	10
Perfluorotridecanoic acid (PFTriA)	<18		18	ng/L		10/16/18 08:36	10/25/18 04:41	10
Perfluorotetradecanoic acid (PFTeA)	<18		18	ng/L		10/16/18 08:36	10/25/18 04:41	10
Perfluorobutanesulfonic acid (PFBS)	1200		18	ng/L		10/16/18 08:36	10/25/18 04:41	10
Perfluoropentanesulfonic acid (PFPeS)	200		18	ng/L		10/16/18 08:36	10/25/18 04:41	10
Perfluorohexanesulfonic acid (PFHxS)	720		18	ng/L		10/16/18 08:36	10/25/18 04:41	10
Perfluoroheptanesulfonic Acid (PFHpS)	<18		18	ng/L		10/16/18 08:36	10/25/18 04:41	10
Perfluorooctanesulfonic acid (PFOS)	140		18	ng/L		10/16/18 08:36	10/25/18 04:41	10
Perfluorononanesulfonic acid (PFNS)	<18		18	ng/L		10/16/18 08:36	10/25/18 04:41	10
Perfluorodecanesulfonic acid (PFDS)	<18		18	ng/L		10/16/18 08:36	10/25/18 04:41	10
Perfluorooctanesulfonamide (FOSA)	<18		18	ng/L		10/16/18 08:36	10/25/18 04:41	10
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	<180		180	ng/L		10/16/18 08:36	10/25/18 04:41	10
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	<180		180	ng/L		10/16/18 08:36	10/25/18 04:41	10
4:2 FTS	<180		180	ng/L		10/16/18 08:36	10/25/18 04:41	10

Isotope Dilution	%Recovery	Qualifier	Limits	Prepared	Analyzed	Dil Fac
13C5 PFPeA	53		25 - 150	10/16/18 08:36	10/25/18 04:41	10
13C2 PFHxA	60		25 - 150	10/16/18 08:36	10/25/18 04:41	10
13C4 PFHpA	79		25 - 150	10/16/18 08:36	10/25/18 04:41	10
13C4 PFOA	88		25 - 150	10/16/18 08:36	10/25/18 04:41	10
13C5 PFNA	91		25 - 150	10/16/18 08:36	10/25/18 04:41	10
13C2 PFDA	102		25 - 150	10/16/18 08:36	10/25/18 04:41	10
13C2 PFUnA	85		25 - 150	10/16/18 08:36	10/25/18 04:41	10
13C2 PFDoA	60		25 - 150	10/16/18 08:36	10/25/18 04:41	10
13C2 PFTeDA	14 *		25 - 150	10/16/18 08:36	10/25/18 04:41	10
13C3 PFBS	85		25 - 150	10/16/18 08:36	10/25/18 04:41	10
18O2 PFHxS	83		25 - 150	10/16/18 08:36	10/25/18 04:41	10
13C4 PFOS	92		25 - 150	10/16/18 08:36	10/25/18 04:41	10
13C8 FOSA	87		25 - 150	10/16/18 08:36	10/25/18 04:41	10
d3-NMeFOSAA	102		25 - 150	10/16/18 08:36	10/25/18 04:41	10
d5-NEtFOSAA	100		25 - 150	10/16/18 08:36	10/25/18 04:41	10

Method: 537 (modified) - Fluorinated Alkyl Substances - DL

Analyte	Result	Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
Perfluorobutanoic acid (PFBA)	840		180	ng/L		10/16/18 08:36	10/25/18 04:34	100
6:2 FTS	<1800		1800	ng/L		10/16/18 08:36	10/25/18 04:34	100
8:2 FTS	<1800		1800	ng/L		10/16/18 08:36	10/25/18 04:34	100

TestAmerica Michigan

Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac

TestAmerica Job ID: 190-17646-1

Client Sample ID: Leachate (duplicate)

Date Collected: 10/03/18 09:35

Date Received: 10/04/18 12:41

Lab Sample ID: 190-17646-2

Matrix: Water

<i>Isotope Dilution</i>	<i>%Recovery</i>	<i>Qualifier</i>	<i>Limits</i>	<i>Prepared</i>	<i>Analyzed</i>	<i>Dil Fac</i>
13C4 PFBA	64		25 - 150	10/16/18 08:36	10/25/18 04:34	100
M2-6:2 FTS	146		25 - 150	10/16/18 08:36	10/25/18 04:34	100
M2-8:2 FTS	94		25 - 150	10/16/18 08:36	10/25/18 04:34	100

Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac

TestAmerica Job ID: 190-17646-1

Client Sample ID: Equipment Blank

Date Collected: 10/03/18 09:25

Date Received: 10/04/18 12:41

Lab Sample ID: 190-17646-3

Matrix: Water

Method: 537 (modified) - Fluorinated Alkyl Substances

Analyte	Result	Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
Perfluorobutanoic acid (PFBA)	<2.4		2.4	ng/L		10/16/18 08:36	10/21/18 03:39	1
Perfluoropentanoic acid (PFPeA)	<2.4		2.4	ng/L		10/16/18 08:36	10/21/18 03:39	1
Perfluorohexanoic acid (PFHxA)	<2.4		2.4	ng/L		10/16/18 08:36	10/21/18 03:39	1
Perfluoroheptanoic acid (PFHpA)	<2.4		2.4	ng/L		10/16/18 08:36	10/21/18 03:39	1
Perfluorooctanoic acid (PFOA)	<2.4		2.4	ng/L		10/16/18 08:36	10/21/18 03:39	1
Perfluorononanoic acid (PFNA)	<2.4		2.4	ng/L		10/16/18 08:36	10/21/18 03:39	1
Perfluorodecanoic acid (PFDA)	<2.4		2.4	ng/L		10/16/18 08:36	10/21/18 03:39	1
Perfluoroundecanoic acid (PFUnA)	<2.4		2.4	ng/L		10/16/18 08:36	10/21/18 03:39	1
Perfluorododecanoic acid (PFDoA)	<2.4		2.4	ng/L		10/16/18 08:36	10/21/18 03:39	1
Perfluorotridecanoic acid (PFTriA)	<2.4		2.4	ng/L		10/16/18 08:36	10/21/18 03:39	1
Perfluorotetradecanoic acid (PFTeA)	<2.4		2.4	ng/L		10/16/18 08:36	10/21/18 03:39	1
Perfluorobutanesulfonic acid (PFBS)	<2.4		2.4	ng/L		10/16/18 08:36	10/21/18 03:39	1
Perfluoropentanesulfonic acid (PFPeS)	<2.4		2.4	ng/L		10/16/18 08:36	10/21/18 03:39	1
Perfluorohexanesulfonic acid (PFHxS)	<2.4		2.4	ng/L		10/16/18 08:36	10/21/18 03:39	1
Perfluoroheptanesulfonic Acid (PFHpS)	<2.4		2.4	ng/L		10/16/18 08:36	10/21/18 03:39	1
Perfluorooctanesulfonic acid (PFOS)	<2.4		2.4	ng/L		10/16/18 08:36	10/21/18 03:39	1
Perfluorononanesulfonic acid (PFNS)	<2.4		2.4	ng/L		10/16/18 08:36	10/21/18 03:39	1
Perfluorodecanesulfonic acid (PFDS)	<2.4		2.4	ng/L		10/16/18 08:36	10/21/18 03:39	1
Perfluorooctanesulfonamide (FOSA)	<2.4		2.4	ng/L		10/16/18 08:36	10/21/18 03:39	1
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	<24		24	ng/L		10/16/18 08:36	10/21/18 03:39	1
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	<24		24	ng/L		10/16/18 08:36	10/21/18 03:39	1
4:2 FTS	<24		24	ng/L		10/16/18 08:36	10/21/18 03:39	1
6:2 FTS	<24		24	ng/L		10/16/18 08:36	10/21/18 03:39	1
8:2 FTS	<24		24	ng/L		10/16/18 08:36	10/21/18 03:39	1
Isotope Dilution	%Recovery	Qualifier	Limits			Prepared	Analyzed	Dil Fac
13C4 PFBA	95		25 - 150			10/16/18 08:36	10/21/18 03:39	1
13C5 PFPeA	96		25 - 150			10/16/18 08:36	10/21/18 03:39	1
13C2 PFHxA	92		25 - 150			10/16/18 08:36	10/21/18 03:39	1
13C4 PFHpA	99		25 - 150			10/16/18 08:36	10/21/18 03:39	1
13C4 PFOA	97		25 - 150			10/16/18 08:36	10/21/18 03:39	1
13C5 PFNA	97		25 - 150			10/16/18 08:36	10/21/18 03:39	1
13C2 PFDA	103		25 - 150			10/16/18 08:36	10/21/18 03:39	1
13C2 PFUnA	97		25 - 150			10/16/18 08:36	10/21/18 03:39	1
13C2 PFDoA	95		25 - 150			10/16/18 08:36	10/21/18 03:39	1
13C2 PFTeDA	94		25 - 150			10/16/18 08:36	10/21/18 03:39	1
13C3 PFBS	94		25 - 150			10/16/18 08:36	10/21/18 03:39	1
18O2 PFHxS	94		25 - 150			10/16/18 08:36	10/21/18 03:39	1
13C4 PFOS	99		25 - 150			10/16/18 08:36	10/21/18 03:39	1
13C8 FOSA	90		25 - 150			10/16/18 08:36	10/21/18 03:39	1
d3-NMeFOSAA	99		25 - 150			10/16/18 08:36	10/21/18 03:39	1
d5-NEtFOSAA	95		25 - 150			10/16/18 08:36	10/21/18 03:39	1
M2-6:2 FTS	105		25 - 150			10/16/18 08:36	10/21/18 03:39	1
M2-8:2 FTS	104		25 - 150			10/16/18 08:36	10/21/18 03:39	1

TestAmerica Michigan

Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac

TestAmerica Job ID: 190-17646-1

Client Sample ID: Field Blank

Date Collected: 10/03/18 00:00

Date Received: 10/04/18 12:41

Lab Sample ID: 190-17646-4

Matrix: Water

Method: 537 (modified) - Fluorinated Alkyl Substances

Analyte	Result	Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
Perfluorobutanoic acid (PFBA)	<1.7		1.7	ng/L		10/16/18 08:36	10/21/18 03:46	1
Perfluoropentanoic acid (PFPeA)	<1.7		1.7	ng/L		10/16/18 08:36	10/21/18 03:46	1
Perfluorohexanoic acid (PFHxA)	<1.7		1.7	ng/L		10/16/18 08:36	10/21/18 03:46	1
Perfluoroheptanoic acid (PFHpA)	<1.7		1.7	ng/L		10/16/18 08:36	10/21/18 03:46	1
Perfluorooctanoic acid (PFOA)	<1.7		1.7	ng/L		10/16/18 08:36	10/21/18 03:46	1
Perfluorononanoic acid (PFNA)	<1.7		1.7	ng/L		10/16/18 08:36	10/21/18 03:46	1
Perfluorodecanoic acid (PFDA)	<1.7		1.7	ng/L		10/16/18 08:36	10/21/18 03:46	1
Perfluoroundecanoic acid (PFUnA)	<1.7		1.7	ng/L		10/16/18 08:36	10/21/18 03:46	1
Perfluorododecanoic acid (PFDoA)	<1.7		1.7	ng/L		10/16/18 08:36	10/21/18 03:46	1
Perfluorotridecanoic acid (PFTriA)	<1.7		1.7	ng/L		10/16/18 08:36	10/21/18 03:46	1
Perfluorotetradecanoic acid (PFTeA)	<1.7		1.7	ng/L		10/16/18 08:36	10/21/18 03:46	1
Perfluorobutanesulfonic acid (PFBS)	<1.7		1.7	ng/L		10/16/18 08:36	10/21/18 03:46	1
Perfluoropentanesulfonic acid (PFPeS)	<1.7		1.7	ng/L		10/16/18 08:36	10/21/18 03:46	1
Perfluorohexanesulfonic acid (PFHxS)	<1.7		1.7	ng/L		10/16/18 08:36	10/21/18 03:46	1
Perfluoroheptanesulfonic Acid (PFHpS)	<1.7		1.7	ng/L		10/16/18 08:36	10/21/18 03:46	1
Perfluorooctanesulfonic acid (PFOS)	<1.7		1.7	ng/L		10/16/18 08:36	10/21/18 03:46	1
Perfluorononanesulfonic acid (PFNS)	<1.7		1.7	ng/L		10/16/18 08:36	10/21/18 03:46	1
Perfluorodecanesulfonic acid (PFDS)	<1.7		1.7	ng/L		10/16/18 08:36	10/21/18 03:46	1
Perfluorooctanesulfonamide (FOSA)	<1.7		1.7	ng/L		10/16/18 08:36	10/21/18 03:46	1
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	<17		17	ng/L		10/16/18 08:36	10/21/18 03:46	1
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	<17		17	ng/L		10/16/18 08:36	10/21/18 03:46	1
4:2 FTS	<17		17	ng/L		10/16/18 08:36	10/21/18 03:46	1
6:2 FTS	<17		17	ng/L		10/16/18 08:36	10/21/18 03:46	1
8:2 FTS	<17		17	ng/L		10/16/18 08:36	10/21/18 03:46	1
Isotope Dilution	%Recovery	Qualifier	Limits			Prepared	Analyzed	Dil Fac
13C4 PFBA	98		25 - 150			10/16/18 08:36	10/21/18 03:46	1
13C5 PFPeA	101		25 - 150			10/16/18 08:36	10/21/18 03:46	1
13C2 PFHxA	99		25 - 150			10/16/18 08:36	10/21/18 03:46	1
13C4 PFHpA	95		25 - 150			10/16/18 08:36	10/21/18 03:46	1
13C4 PFOA	93		25 - 150			10/16/18 08:36	10/21/18 03:46	1
13C5 PFNA	98		25 - 150			10/16/18 08:36	10/21/18 03:46	1
13C2 PFDA	100		25 - 150			10/16/18 08:36	10/21/18 03:46	1
13C2 PFUnA	97		25 - 150			10/16/18 08:36	10/21/18 03:46	1
13C2 PFDoA	92		25 - 150			10/16/18 08:36	10/21/18 03:46	1
13C2 PFTeDA	100		25 - 150			10/16/18 08:36	10/21/18 03:46	1
13C3 PFBS	100		25 - 150			10/16/18 08:36	10/21/18 03:46	1
18O2 PFHxS	93		25 - 150			10/16/18 08:36	10/21/18 03:46	1
13C4 PFOS	100		25 - 150			10/16/18 08:36	10/21/18 03:46	1
13C8 FOSA	93		25 - 150			10/16/18 08:36	10/21/18 03:46	1
d3-NMeFOSAA	98		25 - 150			10/16/18 08:36	10/21/18 03:46	1
d5-NEtFOSAA	98		25 - 150			10/16/18 08:36	10/21/18 03:46	1
M2-6:2 FTS	96		25 - 150			10/16/18 08:36	10/21/18 03:46	1
M2-8:2 FTS	94		25 - 150			10/16/18 08:36	10/21/18 03:46	1

TestAmerica Michigan

QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac

TestAmerica Job ID: 190-17646-1

Method: 537 (modified) - Fluorinated Alkyl Substances

Lab Sample ID: MB 320-252552/1-A

Matrix: Water

Analysis Batch: 253691

Client Sample ID: Method Blank

Prep Type: Total/NA

Prep Batch: 252552

Analyte	MB Result	MB Qualifier	RL	Unit	D	Prepared	Analyzed	Dil Fac
Perfluorobutanoic acid (PFBA)	<2.0		2.0	ng/L		10/16/18 08:36	10/21/18 01:25	1
Perfluoropentanoic acid (PFPeA)	<2.0		2.0	ng/L		10/16/18 08:36	10/21/18 01:25	1
Perfluorohexanoic acid (PFHxA)	<2.0		2.0	ng/L		10/16/18 08:36	10/21/18 01:25	1
Perfluoroheptanoic acid (PFHpA)	<2.0		2.0	ng/L		10/16/18 08:36	10/21/18 01:25	1
Perfluorooctanoic acid (PFOA)	<2.0		2.0	ng/L		10/16/18 08:36	10/21/18 01:25	1
Perfluorononanoic acid (PFNA)	<2.0		2.0	ng/L		10/16/18 08:36	10/21/18 01:25	1
Perfluorodecanoic acid (PFDA)	<2.0		2.0	ng/L		10/16/18 08:36	10/21/18 01:25	1
Perfluoroundecanoic acid (PFUnA)	<2.0		2.0	ng/L		10/16/18 08:36	10/21/18 01:25	1
Perfluorododecanoic acid (PFDoA)	<2.0		2.0	ng/L		10/16/18 08:36	10/21/18 01:25	1
Perfluorotridecanoic acid (PFTriA)	<2.0		2.0	ng/L		10/16/18 08:36	10/21/18 01:25	1
Perfluorotetradecanoic acid (PFTeA)	<2.0		2.0	ng/L		10/16/18 08:36	10/21/18 01:25	1
Perfluorobutanesulfonic acid (PFBS)	<2.0		2.0	ng/L		10/16/18 08:36	10/21/18 01:25	1
Perfluoropentanesulfonic acid (PFPeS)	<2.0		2.0	ng/L		10/16/18 08:36	10/21/18 01:25	1
Perfluorohexanesulfonic acid (PFHxS)	<2.0		2.0	ng/L		10/16/18 08:36	10/21/18 01:25	1
Perfluoroheptanesulfonic Acid (PFHpS)	<2.0		2.0	ng/L		10/16/18 08:36	10/21/18 01:25	1
Perfluorooctanesulfonic acid (PFOS)	<2.0		2.0	ng/L		10/16/18 08:36	10/21/18 01:25	1
Perfluorononanesulfonic acid (PFNS)	<2.0		2.0	ng/L		10/16/18 08:36	10/21/18 01:25	1
Perfluorodecanesulfonic acid (PFDS)	<2.0		2.0	ng/L		10/16/18 08:36	10/21/18 01:25	1
Perfluorooctanesulfonamide (FOSA)	<2.0		2.0	ng/L		10/16/18 08:36	10/21/18 01:25	1
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	<20		20	ng/L		10/16/18 08:36	10/21/18 01:25	1
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	<20		20	ng/L		10/16/18 08:36	10/21/18 01:25	1
4:2 FTS	<20		20	ng/L		10/16/18 08:36	10/21/18 01:25	1
6:2 FTS	<20		20	ng/L		10/16/18 08:36	10/21/18 01:25	1
8:2 FTS	<20		20	ng/L		10/16/18 08:36	10/21/18 01:25	1

Isotope Dilution	MB %Recovery	MB Qualifier	Limits	Prepared	Analyzed	Dil Fac
13C4 PFBA	85		25 - 150	10/16/18 08:36	10/21/18 01:25	1
13C5 PFPeA	89		25 - 150	10/16/18 08:36	10/21/18 01:25	1
13C2 PFHxA	87		25 - 150	10/16/18 08:36	10/21/18 01:25	1
13C4 PFHpA	95		25 - 150	10/16/18 08:36	10/21/18 01:25	1
13C4 PFOA	91		25 - 150	10/16/18 08:36	10/21/18 01:25	1
13C5 PFNA	92		25 - 150	10/16/18 08:36	10/21/18 01:25	1
13C2 PFDA	87		25 - 150	10/16/18 08:36	10/21/18 01:25	1
13C2 PFUnA	91		25 - 150	10/16/18 08:36	10/21/18 01:25	1
13C2 PFDoA	83		25 - 150	10/16/18 08:36	10/21/18 01:25	1
13C2 PFTeA	86		25 - 150	10/16/18 08:36	10/21/18 01:25	1
13C3 PFBS	86		25 - 150	10/16/18 08:36	10/21/18 01:25	1
18O2 PFHxS	89		25 - 150	10/16/18 08:36	10/21/18 01:25	1
13C4 PFOS	90		25 - 150	10/16/18 08:36	10/21/18 01:25	1
13C8 FOSA	84		25 - 150	10/16/18 08:36	10/21/18 01:25	1
d3-NMeFOSAA	89		25 - 150	10/16/18 08:36	10/21/18 01:25	1
d5-NEtFOSAA	85		25 - 150	10/16/18 08:36	10/21/18 01:25	1
M2-6:2 FTS	88		25 - 150	10/16/18 08:36	10/21/18 01:25	1
M2-8:2 FTS	84		25 - 150	10/16/18 08:36	10/21/18 01:25	1

TestAmerica Michigan

QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac

TestAmerica Job ID: 190-17646-1

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: LCS 320-252552/2-A

Matrix: Water

Analysis Batch: 253691

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Prep Batch: 252552

Analyte	Spike Added	LCS Result	LCS Qualifier	Unit	D	%Rec	Limits
Perfluorobutanoic acid (PFBA)	40.0	41.3		ng/L		103	70 - 130
Perfluoropentanoic acid (PFPeA)	40.0	40.3		ng/L		101	66 - 126
Perfluorohexanoic acid (PFHxA)	40.0	38.2		ng/L		95	66 - 126
Perfluoroheptanoic acid (PFHpA)	40.0	38.2		ng/L		95	66 - 126
Perfluorooctanoic acid (PFOA)	40.0	37.6		ng/L		94	64 - 124
Perfluorononanoic acid (PFNA)	40.0	42.6		ng/L		106	68 - 128
Perfluorodecanoic acid (PFDA)	40.0	39.8		ng/L		100	69 - 129
Perfluoroundecanoic acid (PFUnA)	40.0	38.9		ng/L		97	60 - 120
Perfluorododecanoic acid (PFDoA)	40.0	38.7		ng/L		97	71 - 131
Perfluorotridecanoic acid (PFTriA)	40.0	42.3		ng/L		106	72 - 132
Perfluorotetradecanoic acid (PFTeA)	40.0	39.3		ng/L		98	68 - 128
Perfluorobutanesulfonic acid (PFBS)	35.4	32.7		ng/L		92	73 - 133
Perfluoropentanesulfonic acid (PFPeS)	37.5	40.5		ng/L		108	70 - 130
Perfluorohexanesulfonic acid (PFHxS)	36.4	32.8		ng/L		90	63 - 123
Perfluoroheptanesulfonic Acid (PFHpS)	38.1	37.8		ng/L		99	68 - 128
Perfluorooctanesulfonic acid (PFOS)	37.1	34.4		ng/L		93	67 - 127
Perfluorononanesulfonic acid (PFNS)	38.4	36.8		ng/L		96	70 - 130
Perfluorodecanesulfonic acid (PFDS)	38.6	36.8		ng/L		95	68 - 128
Perfluorooctanesulfonamide (FOSA)	40.0	38.0		ng/L		95	70 - 130
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	40.0	34.7		ng/L		87	67 - 127
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	40.0	38.2		ng/L		96	65 - 125
4:2 FTS	37.4	36.3		ng/L		97	70 - 130
6:2 FTS	37.9	32.1		ng/L		85	66 - 126
8:2 FTS	38.3	36.7		ng/L		96	67 - 127

Isotope Dilution	LCS %Recovery	LCS Qualifier	Limits
13C4 PFBA	90		25 - 150
13C5 PFPeA	90		25 - 150
13C2 PFHxA	94		25 - 150
13C4 PFHpA	93		25 - 150
13C4 PFOA	94		25 - 150
13C5 PFNA	90		25 - 150
13C2 PFDA	91		25 - 150
13C2 PFUnA	90		25 - 150
13C2 PFDoA	88		25 - 150
13C2 PFTeDA	96		25 - 150
13C3 PFBS	90		25 - 150

TestAmerica Michigan

QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac

TestAmerica Job ID: 190-17646-1

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: LCS 320-252552/2-A

Matrix: Water

Analysis Batch: 253691

Client Sample ID: Lab Control Sample

Prep Type: Total/NA

Prep Batch: 252552

Isotope Dilution	LCS %Recovery	LCS Qualifier	Limits
18O2 PFHxS	96		25 - 150
13C4 PFOS	98		25 - 150
13C8 FOSA	88		25 - 150
d3-NMeFOSAA	95		25 - 150
d5-NEtFOSAA	95		25 - 150
M2-6:2 FTS	93		25 - 150
M2-8:2 FTS	93		25 - 150

Lab Sample ID: LCSD 320-252552/3-A

Matrix: Water

Analysis Batch: 253691

Client Sample ID: Lab Control Sample Dup

Prep Type: Total/NA

Prep Batch: 252552

Analyte	Spike Added	LCSD Result	LCSD Qualifier	Unit	D	%Rec	%Rec. Limits	RPD	RPD Limit
Perfluorobutanoic acid (PFBA)	40.0	41.3		ng/L		103	70 - 130	0	30
Perfluoropentanoic acid (PFPeA)	40.0	39.4		ng/L		98	66 - 126	2	30
Perfluorohexanoic acid (PFHxA)	40.0	41.5		ng/L		104	66 - 126	8	30
Perfluoroheptanoic acid (PFHpA)	40.0	39.0		ng/L		98	66 - 126	2	30
Perfluorooctanoic acid (PFOA)	40.0	42.3		ng/L		106	64 - 124	12	30
Perfluorononanoic acid (PFNA)	40.0	43.0		ng/L		108	68 - 128	1	30
Perfluorodecanoic acid (PFDA)	40.0	40.8		ng/L		102	69 - 129	2	30
Perfluoroundecanoic acid (PFUnA)	40.0	37.4		ng/L		94	60 - 120	4	30
Perfluorododecanoic acid (PFDoA)	40.0	41.0		ng/L		103	71 - 131	6	30
Perfluorotridecanoic acid (PFTriA)	40.0	43.6		ng/L		109	72 - 132	3	30
Perfluorotetradecanoic acid (PFTeA)	40.0	39.8		ng/L		100	68 - 128	1	30
Perfluorobutanesulfonic acid (PFBS)	35.4	34.0		ng/L		96	73 - 133	4	30
Perfluoropentanesulfonic acid (PFPeS)	37.5	40.4		ng/L		108	70 - 130	0	30
Perfluorohexanesulfonic acid (PFHxS)	36.4	33.5		ng/L		92	63 - 123	2	30
Perfluoroheptanesulfonic Acid (PFHpS)	38.1	37.3		ng/L		98	68 - 128	1	30
Perfluorooctanesulfonic acid (PFOS)	37.1	33.3		ng/L		90	67 - 127	3	30
Perfluorononanesulfonic acid (PFNS)	38.4	36.9		ng/L		96	70 - 130	0	30
Perfluorodecanesulfonic acid (PFDS)	38.6	36.7		ng/L		95	68 - 128	0	30
Perfluorooctanesulfonamide (FOSA)	40.0	38.9		ng/L		97	70 - 130	2	30
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	40.0	36.7		ng/L		92	67 - 127	6	30
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	40.0	37.2		ng/L		93	65 - 125	3	30
4:2 FTS	37.4	36.2		ng/L		97	70 - 130	0	30
6:2 FTS	37.9	34.3		ng/L		90	66 - 126	7	30
8:2 FTS	38.3	39.3		ng/L		103	67 - 127	7	30

TestAmerica Michigan

QC Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac

TestAmerica Job ID: 190-17646-1

<i>Isotope Dilution</i>	<i>LCS D LCS D</i>		<i>Limits</i>
	<i>%Recovery</i>	<i>Qualifier</i>	
13C4 PFBA	97		25 - 150
13C5 PFPeA	98		25 - 150
13C2 PFHxA	93		25 - 150
13C4 PFHpA	97		25 - 150
13C4 PFOA	94		25 - 150
13C5 PFNA	95		25 - 150
13C2 PFDA	97		25 - 150
13C2 PFUnA	100		25 - 150
13C2 PFDoA	88		25 - 150
13C2 PFTeDA	95		25 - 150
13C3 PFBS	98		25 - 150
18O2 PFHxS	99		25 - 150
13C4 PFOS	105		25 - 150
13C8 FOSA	91		25 - 150
d3-NMeFOSAA	96		25 - 150
d5-NEtFOSAA	103		25 - 150
M2-6:2 FTS	96		25 - 150
M2-8:2 FTS	97		25 - 150

Isotope Dilution Summary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac

TestAmerica Job ID: 190-17646-1

Method: 537 (modified) - Fluorinated Alkyl Substances

Matrix: Water

Prep Type: Total/NA

Percent Isotope Dilution Recovery (Acceptance Limits)		
Lab Sample ID	Client Sample ID	M262FTS (25-150)
190-17646-1 - DL	Leachate	114
Surrogate Legend		
M262FTS = M2-6:2 FTS		

Method: 537 (modified) - Fluorinated Alkyl Substances

Matrix: Water

Prep Type: Total/NA

Percent Isotope Dilution Recovery (Acceptance Limits)									
Lab Sample ID	Client Sample ID	PFBA (25-150)	PFPeA (25-150)	PFHxA (25-150)	PFHpA (25-150)	PFOA (25-150)	PFNA (25-150)	PFDA (25-150)	PFUnA (25-150)
190-17646-1	Leachate	27	58	62	81	92	91	92	85
190-17646-2 - DL	Leachate (duplicate)	64							
190-17646-2	Leachate (duplicate)		53	60	79	88	91	102	85
190-17646-3	Equipment Blank	95	96	92	99	97	97	103	97
190-17646-4	Field Blank	98	101	99	95	93	98	100	97
LCS 320-252552/2-A	Lab Control Sample	90	90	94	93	94	90	91	90
LCSD 320-252552/3-A	Lab Control Sample Dup	97	98	93	97	94	95	97	100
MB 320-252552/1-A	Method Blank	85	89	87	95	91	92	87	91
Percent Isotope Dilution Recovery (Acceptance Limits)									
Lab Sample ID	Client Sample ID	PFDaA (25-150)	PFTDA (25-150)	3C3-PFBs (25-150)	PFHxS (25-150)	PFOS (25-150)	PFOSA (25-150)	NMeFOS (25-150)	NEtFOS (25-150)
190-17646-1	Leachate	55	15 *	108	84	91	80	92	91
190-17646-2 - DL	Leachate (duplicate)								
190-17646-2	Leachate (duplicate)	60	14 *	85	83	92	87	102	100
190-17646-3	Equipment Blank	95	94	94	94	99	90	99	95
190-17646-4	Field Blank	92	100	100	93	100	93	98	98
LCS 320-252552/2-A	Lab Control Sample	88	96	90	96	98	88	95	95
LCSD 320-252552/3-A	Lab Control Sample Dup	88	95	98	99	105	91	96	103
MB 320-252552/1-A	Method Blank	83	86	86	89	90	84	89	85
Percent Isotope Dilution Recovery (Acceptance Limits)									
Lab Sample ID	Client Sample ID	M262FTS (25-150)	M282FTS (25-150)						
190-17646-1	Leachate		132						
190-17646-2 - DL	Leachate (duplicate)	146	94						
190-17646-2	Leachate (duplicate)								
190-17646-3	Equipment Blank	105	104						
190-17646-4	Field Blank	96	94						
LCS 320-252552/2-A	Lab Control Sample	93	93						
LCSD 320-252552/3-A	Lab Control Sample Dup	96	97						
MB 320-252552/1-A	Method Blank	88	84						
Surrogate Legend									
PFBA = 13C4 PFBA									
PFPeA = 13C5 PFPeA									
PFHxA = 13C2 PFHxA									
PFHpA = 13C4 PFHpA									
PFOA = 13C4 PFOA									
PFNA = 13C5 PFNA									

TestAmerica Michigan

Isotope Dilution Summary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac

TestAmerica Job ID: 190-17646-1

PFDA = 13C2 PFDA
PFUnA = 13C2 PFUnA
PFDoA = 13C2 PFDoA
PFTDA = 13C2 PFTeDA
13C3-PFBS = 13C3 PFBS
PFHxS = 18O2 PFHxS
PFOS = 13C4 PFOS
PFOSA = 13C8 FOSA
d3-NMeFOSAA = d3-NMeFOSAA
d5-NEtFOSAA = d5-NEtFOSAA
M262FTS = M2-6:2 FTS
M282FTS = M2-8:2 FTS

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Definitions/Glossary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac

TestAmerica Job ID: 190-17646-1

Qualifiers

LCMS

Qualifier	Qualifier Description
*	Isotope Dilution analyte is outside acceptance limits.

Glossary

Abbreviation	These commonly used abbreviations may or may not be present in this report.
α	Listed under the "D" column to designate that the result is reported on a dry weight basis
%R	Percent Recovery
CFL	Contains Free Liquid
CNF	Contains No Free Liquid
DER	Duplicate Error Ratio (normalized absolute difference)
Dil Fac	Dilution Factor
DL	Detection Limit (DoD/DOE)
DL, RA, RE, IN	Indicates a Dilution, Re-analysis, Re-extraction, or additional Initial metals/anion analysis of the sample
DLC	Decision Level Concentration (Radiochemistry)
EDL	Estimated Detection Limit (Dioxin)
LOD	Limit of Detection (DoD/DOE)
LOQ	Limit of Quantitation (DoD/DOE)
MDA	Minimum Detectable Activity (Radiochemistry)
MDC	Minimum Detectable Concentration (Radiochemistry)
MDL	Method Detection Limit
ML	Minimum Level (Dioxin)
NC	Not Calculated
ND	Not Detected at the reporting limit (or MDL or EDL if shown)
PQL	Practical Quantitation Limit
QC	Quality Control
RER	Relative Error Ratio (Radiochemistry)
RL	Reporting Limit or Requested Limit (Radiochemistry)
RPD	Relative Percent Difference, a measure of the relative difference between two points
TEF	Toxicity Equivalent Factor (Dioxin)
TEQ	Toxicity Equivalent Quotient (Dioxin)

QC Association Summary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac

TestAmerica Job ID: 190-17646-1

LCMS

Prep Batch: 252552

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
190-17646-1 - DL	Leachate	Total/NA	Water	3535	
190-17646-1	Leachate	Total/NA	Water	3535	
190-17646-2 - DL	Leachate (duplicate)	Total/NA	Water	3535	
190-17646-2	Leachate (duplicate)	Total/NA	Water	3535	
190-17646-3	Equipment Blank	Total/NA	Water	3535	
190-17646-4	Field Blank	Total/NA	Water	3535	
MB 320-252552/1-A	Method Blank	Total/NA	Water	3535	
LCS 320-252552/2-A	Lab Control Sample	Total/NA	Water	3535	
LCSD 320-252552/3-A	Lab Control Sample Dup	Total/NA	Water	3535	

Analysis Batch: 253691

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
190-17646-3	Equipment Blank	Total/NA	Water	537 (modified)	252552
190-17646-4	Field Blank	Total/NA	Water	537 (modified)	252552
MB 320-252552/1-A	Method Blank	Total/NA	Water	537 (modified)	252552
LCS 320-252552/2-A	Lab Control Sample	Total/NA	Water	537 (modified)	252552
LCSD 320-252552/3-A	Lab Control Sample Dup	Total/NA	Water	537 (modified)	252552

Analysis Batch: 254698

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
190-17646-1 - DL	Leachate	Total/NA	Water	537 (modified)	252552
190-17646-1	Leachate	Total/NA	Water	537 (modified)	252552
190-17646-2 - DL	Leachate (duplicate)	Total/NA	Water	537 (modified)	252552
190-17646-2	Leachate (duplicate)	Total/NA	Water	537 (modified)	252552

Lab Chronicle

Client: City of Cadillac Utilities
Project/Site: City of Cadillac

TestAmerica Job ID: 190-17646-1

Client Sample ID: Leachate

Date Collected: 10/03/18 09:30

Date Received: 10/04/18 12:41

Lab Sample ID: 190-17646-1

Matrix: Water

Prep Type	Batch Type	Batch Method	Run	Dilution Factor	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	3535	DL		252552	10/16/18 08:36	MYV	TAL SAC
Total/NA	Analysis	537 (modified)	DL	100	254698	10/25/18 04:19	JRB	TAL SAC
Total/NA	Prep	3535			252552	10/16/18 08:36	MYV	TAL SAC
Total/NA	Analysis	537 (modified)		10	254698	10/25/18 04:26	JRB	TAL SAC

Client Sample ID: Leachate (duplicate)

Date Collected: 10/03/18 09:35

Date Received: 10/04/18 12:41

Lab Sample ID: 190-17646-2

Matrix: Water

Prep Type	Batch Type	Batch Method	Run	Dilution Factor	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	3535	DL		252552	10/16/18 08:36	MYV	TAL SAC
Total/NA	Analysis	537 (modified)	DL	100	254698	10/25/18 04:34	JRB	TAL SAC
Total/NA	Prep	3535			252552	10/16/18 08:36	MYV	TAL SAC
Total/NA	Analysis	537 (modified)		10	254698	10/25/18 04:41	JRB	TAL SAC

Client Sample ID: Equipment Blank

Date Collected: 10/03/18 09:25

Date Received: 10/04/18 12:41

Lab Sample ID: 190-17646-3

Matrix: Water

Prep Type	Batch Type	Batch Method	Run	Dilution Factor	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	3535			252552	10/16/18 08:36	MYV	TAL SAC
Total/NA	Analysis	537 (modified)		1	253691	10/21/18 03:39	AAR	TAL SAC

Client Sample ID: Field Blank

Date Collected: 10/03/18 00:00

Date Received: 10/04/18 12:41

Lab Sample ID: 190-17646-4

Matrix: Water

Prep Type	Batch Type	Batch Method	Run	Dilution Factor	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	3535			252552	10/16/18 08:36	MYV	TAL SAC
Total/NA	Analysis	537 (modified)		1	253691	10/21/18 03:46	AAR	TAL SAC

Laboratory References:

TAL SAC = TestAmerica Sacramento, 880 Riverside Parkway, West Sacramento, CA 95605, TEL (916)373-5600

Analyst References:

Lab: TAL SAC

Batch Type: Prep

MYV = Mai Yee Vang

Batch Type: Analysis

AAR = Amani Royce

JRB = John Barnett

TestAmerica Michigan

Accreditation/Certification Summary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac

TestAmerica Job ID: 190-17646-1

Laboratory: TestAmerica Michigan

All accreditations/certifications held by this laboratory are listed. Not all accreditations/certifications are applicable to this report.

Authority	Program	EPA Region	Identification Number	Expiration Date
Michigan	State Program	5	57	05-05-20

Laboratory: TestAmerica Sacramento

All accreditations/certifications held by this laboratory are listed. Not all accreditations/certifications are applicable to this report.

Authority	Program	EPA Region	Identification Number	Expiration Date
Alaska (UST)	State Program	10	17-020	01-20-21
ANAB	DoD ELAP		L2468	01-20-21
Arizona	State Program	9	AZ0708	08-11-19
Arkansas DEQ	State Program	6	88-0691	06-17-19
California	State Program	9	2897	01-31-19
Colorado	State Program	8	CA00044	08-31-19
Connecticut	State Program	1	PH-0691	06-30-19
Florida	NELAP	4	E87570	06-30-19
Georgia	State Program	4	N/A	01-28-19
Hawaii	State Program	9	N/A	01-29-19
Illinois	NELAP	5	200060	03-17-19
Kansas	NELAP	7	E-10375	10-31-18 *
Louisiana	NELAP	6	30612	06-30-19
Maine	State Program	1	CA0004	04-14-20
Michigan	State Program	5	9947	01-31-20
Nevada	State Program	9	CA00044	07-31-19
New Hampshire	NELAP	1	2997	04-18-19
New Jersey	NELAP	2	CA005	06-30-19
New York	NELAP	2	11666	03-31-19
Oregon	NELAP	10	4040	01-29-19
Pennsylvania	NELAP	3	68-01272	03-31-19
Texas	NELAP	6	T104704399	05-31-19
US Fish & Wildlife	Federal		LE148388-0	07-31-19
USDA	Federal		P330-18-00239	01-17-21
USEPA UCMR	Federal	1	CA00044	12-31-20
Utah	NELAP	8	CA00044	02-28-19
Vermont	State Program	1	VT-4040	04-30-19
Virginia	NELAP	3	460278	03-14-19
Washington	State Program	10	C581	05-05-19
West Virginia (DW)	State Program	3	9930C	12-31-18
Wyoming	State Program	8	8TMS-L	01-28-19

* Accreditation/Certification renewal pending - accreditation/certification considered valid.

TestAmerica Michigan

Chain of Custody Record

[illegible]

☐ MSDS or Known Hazard Information Supplied by Client
☐ Bottle stickers applied ☐ ELEMENT comment entered ☐ MSDS COC scanned & mailed to EH&S
☐ Discrepancies Client ID City of Cadillac
☐ Short Hold Work Order # 190-17646
☐ Rush ☐ 24hr ☐ 2day ☐ 3day ☐ 5day ☐ Other
 Receipt evaluation performed by - Initials AMY Date 10/4/18 Time 12:41

Cooler/Sample Receipt

(AFTER HOURS receipt, complete gray areas)
 Place cooler in walk-in, place this form in Receiving in-
 box Date Time rec'd Initials

Method of Shipment:

☐ Walk-In Client ☐ TestAmerica Field/Courier
☐ Other Client/3rd Party Courier
☐ Fed Ex Tracking #
☒ UPS Tracking # 1Z547 491 03 5470 1423
☐ Other Ground

Shipping Container Type:

☐ Cooler ☒ Box Insulated
☐ None ☒ Other Sh. Affair
Packing Materials:
☒ Plastic Bags ☐ Foam
☐ Bubble Wrap ☐ Paper
☐ Packing Peanuts ☐ None
☐ Other

Custody Seals Intact:

☐ Yes ☐ No
☒ N/A (not used or required)
Cooling Materials:
☒ Ice (solid) ☐ Ice (Melted)
☐ Blue Ice ☐ None
☐ Other

Bacteriological Temp (°C) Corrected
 Samples

Frozen
 yes no

Received within 2 hours
 yes no

Sample Flagged
 yes no

Receipt Temperatures

Thermometer ID Observed (°C) Corrected (°C) Blank Temp Sample Temp
140252483 0.0 0.0 ☐ ☒
140252476 0.0 0.0 ☐ ☒
☐ ☐ ☐ ☐
☐ ☐ ☐ ☐
☐ ☐ ☐ ☐

Received on

☐ Check if Additional Sheets Required

* Receipt temperatures are considered acceptable if the samples are received on the same day they were collected & show signs that the cooling process has started. Temperature acceptance for most tests is ≤6.0°C, but not frozen. For additional information, please refer to SOP DT-SCA-004 Sample Receipt and Login, Attachment 2 - Holding Times, Preservation and Container Requirements

Receipt Questions**

	Y	N	n/a	"No" answers require additional comment
COC present & TA receipt signature, date, & time properly documented?	✓			
Containers & labels in good condition? (unbroken, not leaking, appropriately filled, labels legible & attached)	✓			
Appropriate containers used & adequate volume provided?	✓			
Number of sample containers match COC?	✓			
Samples received within hold time?	✓			
Samples submitted for GRO and Volatiles analyses (8260, 624, 524) received without headspace?			✓	
Was a Trip Blank received with VOA samples?			✓	
Were the samples free of any questionable physical conformities? For example, field duplicates or multiple bottles of the same sample do not significantly vary in appearance (color, proportion of solids, etc.)	✓			
Were the COC, bottle labels, and all other items free of all other discrepancies or issues that would need to be addressed with the Project Manager and/or Client?	✓			

** May not be applicable if samples are not for compliance testing

* Excludes FOG, Volatiles, TOC Vials

Client Contact Record

Contact via: ☐ Phone ☐ Email ☐ Other Person Contacted: Date/Time:
☐ Discrepancy allowance agreement is on record in the client project file
 Discussion/Resolution:

Any additional documentation and clarification from client must be noted in the narrative and/or scanned into the COC directory

Reviewed by PM Signature

Date

WI Page 1 of 1

WI No DT-SCA-WI-001 TO effective 06/11/12

Chain of Custody Record

[illegible]